Unit: 6 Unit 2: Exponents Radicals
Topic: Radicals

Name: ________________
Date: ____________

Quiz (Carolina Reaper)

Q1. A shipping container has a volume of $8 \cdot 8 \cdot 8$ cubic meters. What is the length of one side in meters, expressed as a radical?
(A) $2\sqrt{2}$
(B) $2\sqrt[3]{2}$
(C) 2
(D) $2\sqrt{4}$
(E) 4

Q2. Simplify the radical expression: $\sqrt[3]{27}$
(A) 3
(B) $3\sqrt{3}$
(C) 9
(D) $9\sqrt{3}$
(E) 27

Q3. Traffic flow is measured in the number of cars that pass a point in one hour. If 16 cars pass per hour, what is the number of cars that pass per hour squared, expressed as a radical?
(A) $\sqrt{256}$
(B) $\sqrt[3]{4096}$
(C) 16
(D) $16 \cdot 16$
(E) 4

Q4. A delivery truck travels x kilometers in $\sqrt{x}$ hours. If the truck travels 16 kilometers, how many hours does it take, expressed in terms of x?
(A) $x = 16$
(B) $x = 4$
(C) $x = \sqrt{16}$
(D) $x = 16 \cdot 4$
(E) $x = 2$

Q5. A package is delivered in t days, where t is given by the equation $t = \sqrt[3]{x}$. If the package is delivered in 2 days, what is the value of x?
(A) $x = 8$
(B) $x = 4$
(C) $x = 2$
(D) $x = 1$
(E) $x = 16$

Q6. The time it takes for a truck to travel from one city to another is given by $t = \frac{d}{r}$, where d is the distance and r is the rate. If $d = 24$ kilometers and $r = \sqrt{4}$ kilometers per hour, what is the time in hours?
(A) $t = 6$
(B) $t = 12$
(C) $t = 8$
(D) $t = 4$
(E) $t = 3$

Q7. A ship travels $8 \cdot 8$ kilometers in $\sqrt{64}$ hours. How many kilometers does it travel per hour?
(A) 8 kilometers per hour
(B) 4 kilometers per hour
(C) 16 kilometers per hour
(D) 2 kilometers per hour
(E) 1 kilometer per hour

Q8. A package delivery company uses the equation $d = rt$, where d is the distance, r is the rate, and t is the time. If $d = 12$ kilometers and $t = \sqrt{4}$ hours, what is the rate r?
(A) $r = 3$ kilometers per hour
(B) $r = 6$ kilometers per hour
(C) $r = 12$ kilometers per hour
(D) $r = 2$ kilometers per hour
(E) $r = 4$ kilometers per hour

Open Ended Questions (FRQ)

Q9. A logistics company has two warehouses, one located 8 kilometers from the city center and the other located 27 kilometers from the city center. If a truck travels from the first warehouse to the city center and then to the second warehouse, what is the total distance traveled by the truck in terms of radicals? Show all work and simplify the expression.

Q10. A transportation company uses the equation $t = \frac{d}{\sqrt{r}}$ to determine the time it takes to travel a certain distance d at a rate r. If the company wants to travel 16 kilometers in 2 hours, what is the rate r in terms of radicals? Show all work and simplify the expression.

Chef's Tips

- Double-check the units of measurement in each question.
- Use a calculator only when necessary.
- Show all work and simplify expressions.